

TAHA BIN YOUSUF

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SUMMARY

Software Engineering student (COMSATS University, 6th semester) with a robust full-stack development profile across web, mobile, and desktop environments, combined with high-performance parallel computing and machine learning experience. Developed automated workflows, dynamic e-commerce pipelines, and co-founded an AI-driven virtual try-on application. Seeking a Software Engineering or DS/ML internship to contribute to modern full-stack development, architectural optimization, and product engineering.

EDUCATION

COMSATS University Islamabad, Lahore Campus

Sep 2023 – Present

BSc Software Engineering — 6th Semester

Relevant Coursework: Machine Learning · Parallel & Distributed Computing · Data Structures & Algorithms · Object-Oriented Programming · Software Re-engineering · Database Systems

TECHNICAL SKILLS

Languages:	Python, Java, Dart, JavaScript, C++, SQL
Web & Mobile:	Next.js, React.js, Node.js, Express.js, Flutter, Firebase
ML / Parallel / DB:	scikit-learn, OpenMP, NumPy, Pandas, MongoDB, MySQL
Tools:	Git, GitHub, VS Code, Android Studio, Jupyter Notebook, PyCharm

PROJECTS

Diva.ai – AI-Driven Bridal Assistance Platform

Machine Learning · Computer Vision · Full-Stack

Final Year Project (FYP) — Co-Founder

- Co-founded and architected an AI-powered e-commerce and virtual styling platform engineered for the South Asian bridal retail sector.
- Engineered and evaluated virtual try-on pipelines leveraging Computer Vision and deep learning to simulate garment fits in real time.
- Directed agile sprint cycles, managed complex code integration, and authored the initial system architecture specifications.

Parallel Linear Regression Model with GUI

C++ · OpenMP · Performance Optimization

Parallel & Distributed Computing Project

- Designed a high-performance ML suite to accelerate model training on large multi-dimensional matrices.
- Parallelized cost calculations and batch gradient descent using OpenMP multi-threading, achieving significant speedups over sequential baselines.
- Built an interactive GUI with dynamic graphs to monitor parameter updates and runtime metrics in real time.

Fitness Club Full-Stack Application

Next.js · Node.js · MongoDB · JavaScript

Collaborative Web Project

- Migrated server-side services from a legacy Express repository into the Next.js App Router, improving rendering via React Server Components.
- Designed Mongoose schema layers for scalable data persistence and built custom API routes for robust state validation.
- Managed version control and deployment pipelines through Vercel for seamless CI/CD.

GreenLeaf – Online Nursery Management System

MERN Stack · REST API · Web Development

Academic Web Development Project

- Built a multi-tier web application streamlining retail and inventory workflows for organic nurseries.
- Developed reusable React hooks for front-end state management and authored clean, secured REST API endpoints.

CozyNest – Mobile E-Commerce Application

Flutter · Dart · Firebase Cloud

Personal Portfolio Project

- Engineered a responsive mobile shopping app for home and travel utilities using Flutter.
- Integrated Firebase Cloud for persistent state, real-time product queries, and secure authentication.